Networks

René Serral-Gracià[1]

[1]Universitat Politècnica de Catalunya (UPC)

April 20, 2026

Lectures

1. System administration introduction
2. Operating System installation
3. User management
4. Application management
5. System monitoring
6. Filesystem Maintenance
7. **Network services**
8. Security and Protection
9. Introduction to Public Cloud

Outline

Goals

Knowledge

- Main services and networking protocols
  - Superserver, portmapper, DNS, FTP, WWW, e-mail

Abilities

- Service configurations
  - Superserver
  - DNS
  - FTP
  - WWW
  - E-Mail

Network admin considerations (I)

Security measures

- Never execute services with superuser privileges
- Expose only necessary services – firewalls
- Configure carefully all the offered services
  - Never leave default configurations
  - Disable/Remove unused services
- Monitor the service's logs
- Check for security issues – **be up to date**

Network admin considerations (and II)

Port classification

- Privileged ports: 0 - 1023
  - Controlled and assigned by IANA
  - Only privileged users (`root`) mai install services to those ports
- Registered ports: 1024 - 49151
  - Not controlled but registered by IANA
  - Registry about services using those ports – `/etc/services`
- Dynamic ports: 49152 - 65535
  - Used for temporary connections

/etc/services

- Relates services with corresponding port number
  - various applications use it (netstat, ...)

```
servicename     port/protocol     alias list
```

```
echo            7/tcp
echo            7/udp
systat          11/tcp            users
systat          11/udp            users
ftp-data        20/tcp
ftp-data        20/udp
ftp             21/tcp
ftp             21/udp            fsp fspd
ssh             22/tcp
ssh             22/udp
...
smtp            25/tcp            mail
smtp            25/udp            mail
http            80/tcp            www www-http
http            80/udp            www www-http
```

Network Address Translation – NAT

- Router translates internal addresses by one (or various) of its own
  - Allows using a reserved IP (pool) and keep connectivity to the outside
- The router remembers the output connections to identify its answers
  - Output connection:
    - 192.168.1.25 (port 1085) → 212.106.192.142 (1085)
  - Reply connection:
    - 212.106.192.142 (1085) → 192.168.1.25 (1085)

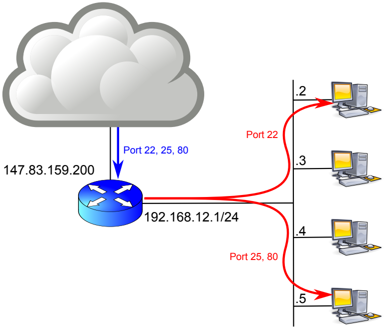
Tools: `iptables` (SNAT, MASQUERADE), `dnsmasq`

NAT collateral effects

- Private addresses are not visible from the outside
  - Attacks may only fall to the router – except over ongoing connections
- Network security depednds on router security
- Internal machines cannot offer services to the outside
  - Except when using Destination Network Address Translation (DNAT)
- Important performance penalty for the network
  - All external connections go through a single router
  - Each packet requires some CPU time for processing
- Some services do not behave properly when using NAT
  - Those establishing connections to the inside
  - FTP, IRC, Netmeeting, . . .

Destination Address Translation (DNAT)

- Indicate to the NAT router it must forward some input connections to a particular machine
- Map router ports to some internal machine

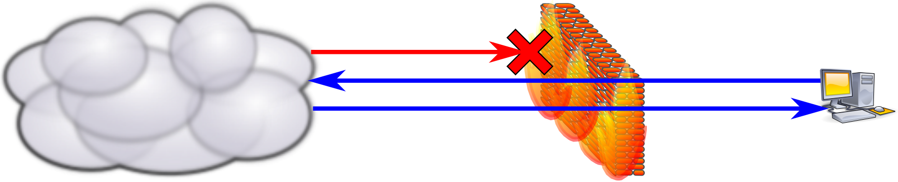


Eines: `iptables` (DNAT)

Firewall

Server that determines which connections may be established between two networks

- It typically works at network and transport layers
  - In general application details are not known
- It can keep connection status (Connection Tracking)
  - It allows related connections: “replies“



Firewall == Security?

- A firewall is another piece of the overall security of a system
- Its use can potentially offer a false sense of security
- Other aspects cannot be neglected
  - Correct application configuration
  - Perform regular security updates on installed software
  - Limit concurrent connections
- Other security tools in the private network and servers are still necessary

Outline

Server types

Connection oriented

- The server keeps status about the different sessions
- Better performance
- Less error resilience

Connectionless

- There is no status about the client connections
- There are no sessions
- Requests must be self contained
- Client request must contain all the required information
- Better failure resilience and recovery

Server types – Depending on authority 1/2

- Primary
  - They keep a copy of all the information
  - If there is mismatch in the stored information the primary takes precedence
  - There is one per service
- Secondary – *Mostly deprecated*
  - Keep copies of the information
  - Performing periodic updates with the primary
  - There can be more than one per service
  - Load balancing
  - Are an implicit backup of the primary

Server types – Depending on authority 2/2

- Cache – *Mostly deprecated*
  - Intercept and **terminate** TCP connections
  - Keep *partial* copies of the most used information
  - It is possible to have more than one per service
    - Better performance
- Proxy – Sometimes called WAF
  - Intercept and **terminate** TCP connections
  - It is possible to have more than one per service
  - They can add **security checks**, filtering, log, . . .
  - Allow connection rerouting
  - Smart balancing

Outline

Superserver

- A service even when idle uses resources
  - Many services are requested only from time to time: `telnet`, `ftp`, `ssh`, ...
- Superserver listens to all the ports and activates the service only when needed
  - It detects the request
  - Initiates the service
  - Passes the message
- Limitations
  - Between connections it is not possible to keep information in memory
  - Overhead caused by process creation

Implementations: `inetd`, `xinetd`

/etc/xinetd.conf, /etc/xinetd.d

Indicates the services offered by the superserver

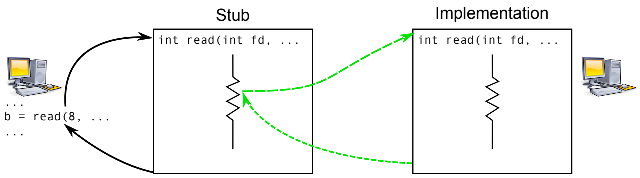
Service, Protocol, User/group, Server, Parameters

```
$ cat /etc/xinetd.conf
includedir /etc/xinetd.d
```

```
$ cat /etc/xinetd.d/ftp
service ftp
{
        socket_type             = stream
        wait                    = no
        user                    = root
        server                  = /usr/sbin/vsftpd
        log_on_success         += HOST DURATION
        log_on_failure         += HOST
        disable                 = no
}
```

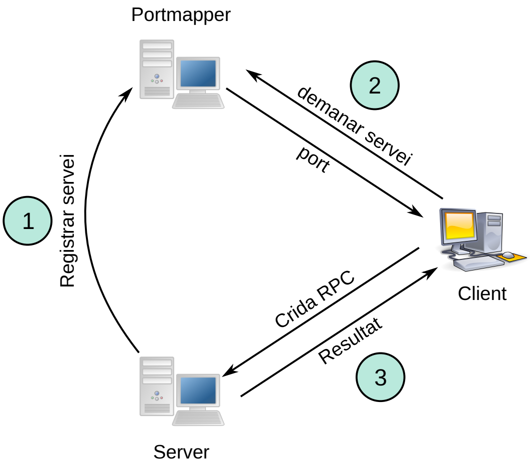
Remote Procedure Calls (RPC)

- Remote subroutine invokation
  - Identified by a service number ID
- RPC Servers
  - They implement a set of remote connections
  - Listen in a dynamic port
- Portmapper
- Registers the RPC servers
  - Maps the port with the subroutines
- Needed by other services
  - NFS, ...



Portmapper

- All the status is kept on memory
  - If the process fails, is not enough restarting it
  - All RPC servers must be restarted
- All services must be registered upon portmapper start



Outline

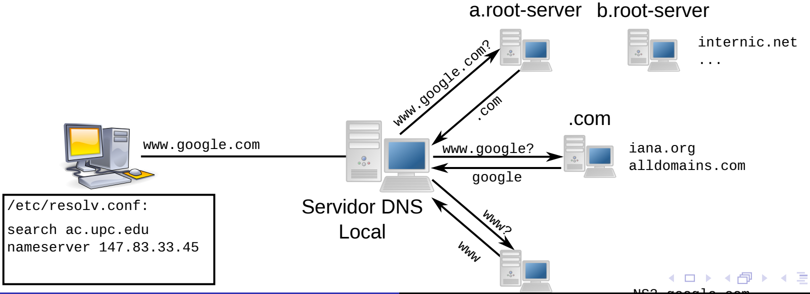
Domain Name System (DNS)

- Name resolution service
  - Hostname → IP address
  - IP Address → hostname
- Issues
  - Large amount of machines
  - Large number of changes
- Solution
  - Hierarchical distribution of the information (domains)
  - Authority delegation

DNS Internals

Authority delegation

- Each domain administers its own server
- Everybody knows the higher servers in the hierarchy (root)
- Everybody knows the server for their domain
- Name resolution is iterative



Service performance

Using ”caches“ is convenient

- High temporal locality
  - Less queries
- High spacial locality
  - Less load on root
  - Avoids some steps

DNS can be used for load balancing

- We can have several IPs for the same name
- Each query returns different values
  - Round Robin
  - Geographical
  - Server load

```
$ nslookup www.google.com
Name:    www.google.com
Address: 212.106.221.23
Name:    www.google.com
Address: 212.106.221.27
...
```

DNS client configuration

- `/etc/host.conf`
  - Where a name is searched and its order
- `/etc/hosts`
  - Locally translated machines
- `/etc/resolv.conf`
  - Automatic domains to be searched
  - IP addresses of the DNS servers

DNS Server configuration

- `/etc/bind/named.conf`
  - What are we administering?
    - DNS Domains
    - IP addresses ranges
  - Indicates primary, secondary, or cache
- Direct translation files
  - Name.domain $\rightarrow$ IP address
  - 1 file for each administered domain
- Inverse translation file
  - IP Address $\rightarrow$ name.domain
  - 1 file for each administered IP range

DNS type of registers

- SOA (Start of Authority)
  - Serial number
  - Refresh time and retries
  - Expiration times
  - Minimum TTL
- A - Direct translation
  - Name → IP address

```
romeu IN A 147.83.32.4
```

- CNAME - synonyms
  - Name → name

```
romeu IN CNAME lp_romeu
```

DNS type of registers

- PTR - inverse translation
  - IP Address → DNS name

```
4 IN PTR romeu.ac.upc.edu.
```

- NS - Domain delegation
  - DNS Domain→ server

```
ac IN NS 147.83.32.3
```

- MX - mail exchanger
  - DNS Domain → server

```
ac IN MX 147.83.33.10
```

- And others. . .
  - HINFO, WKS, . . .

DNS configuration example

Zone "cluster.mygroup.upc.edu", as primary.

```
$ cat /etc/bind/named.conf
options {
        directory "/var/cache/bind";
        forwarders {
                147.83.159.217;
        };
        auth-nxdomain no;    # conform to RFC1035
        listen-on-v6 { any; };
};
zone "cluster.craax.upc.edu"  {
  type master;
  file "/etc/bind/cluster.zone";
};

zone "1.1.10.in-addr.arpa"  {
  type master;
  file "/etc/bind/cluster.rev";
};
```

DNS configuration example

```
$ cat /etc/bind/cluster.zone
$TTL    604800
@       IN      SOA     cluster. cluster.craax.upc.edu. (
                       20101220         ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
;
@       IN      NS      gandalf
$ORIGIN                 cluster.craax.upc.edu.
gandalf         IN      A       10.1.1.1
```

```
$ cat /etc/bind/cluster.rev
$TTL    604800
@       IN      SOA     cluster. cluster.craax.upc.edu. (
                       20101220         ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
;
@       IN      NS      gandalf
$ORIGIN                 cluster.craax.upc.edu.
1       IN      PTR     gandalf.cluster.craax.upc.edu.
```

Exercise

- We have 3 services at (`server1`, `server2` i `server3`) with these registers

```
server1 IN A 123.123.123.1
server2 IN A 123.123.123.2
server3 IN A 123.123.123.3
```

- We want to add the following services
  - `www` at `server1` (`server2` is the backup server)
  - `ftp` at `server1` and `server2`
  - incoming/outgoing mail at `server3`

Which registries would you add?

DNS Related tools

- `whois domain`
  - Provides contact information for a domain
- `dig [@server] query`
  - Performs a DNS query
  - It allows controlling different resources
    - Server, type of register, iterative/recursive resolution, . . .
  - Returns the registers corresponding to the query
    - It supports debugging

Dynamic Host Configuration Protocol (DHCP)

- It delivers automatically the network configuraiton to a host
  - IP allocation
  - Gateway
  - DNS
- Machine trustfulness is not verified
  - By default it is assumed that if the host can reach connectivity then it is legitimate
  - It can provide MAC address verification
- IP addresses are assigned from a predefined range

Dynamic Host Configuration Protocol (DHCP)

Remote boot support through BOOTP and PXE

- Preboot Execution Environment (PXE)
- Network card uses BIOS to get network information
- It allows to decide the kernel image to boot
  - Downloaded through TFTP
  - A remote root system can be mounted

Dynamic Host Configuration Protocol (DHCP)

For /etc/resolv.conf

For PXE

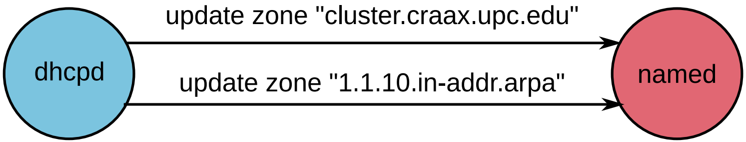
For ip address

For ip route

```
ddns-update-style none;
option domain-name-servers 192.168.1.1;

allow booting;
allow bootp;
default-lease-time 600;
max-lease-time 7200;
authoritative;

subnet 192.168.1.0 netmask 255.255.255.0 {
 range dynamic-bootp 192.168.1.172 192.168.1.254;
 range 192.168.1.2 192.168.1.171;
 filename "pxelinux.0";

 option subnet-mask 255.255.255.0;
 option broadcast-address 192.168.1.255;
 option routers 192.168.1.1;
}
```

Dynamic Host Configuration (DHCP)

DHCP and DNS can work together



/etc/dhcpd/dhcpd.conf

```
ddns-update-style interim;
key DHCP_UPDATER {
   algorithm HMAC-MD5.SIG-ALG.REG.INT;
   secret pRP5FapFoJ95JEL06sv4PQ==;
};
zone ac.upc.edu. {
   primary 192.168.1.1;
   key DHCP_UPDATER;
}
```

/etc/bind/named.conf

```
key DHCP_UPDATER {
   algorithm HMAC-MD5.SIG-ALG.REG.INT;
   secret pRP5FapFoJ95JEL06sv4PQ==;
};
zone ac.upc.edu. {
   type master;
   file "ac.zone";
   allow-update { key DHCP_UPDATER; };
};
...
```

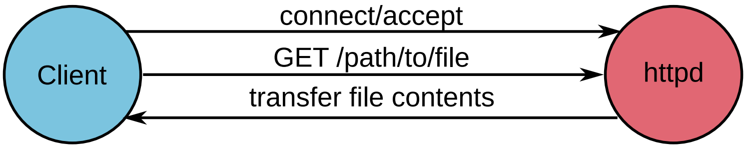
Exercise

In group

- Which potential problem can be caused by a DHCP server crash?
- Propose an implementation to solve it

Hypertext Transfer Protocol (HTTP)

- Data transfer service
- It provides Statelessness, where every request is independent
- It may use TCP
- But in newer implementations is common to use QUIC
- In distributed apps, HTTP is the **Inter-Process Communication (IPC)** mechanism



HTTP Verbs

- Is the way that HTTP indicates the action to be taken
- In RESTful architectures, the method (verb) defines the operation and dictates how the infrastructure (CDN, Proxy, Cache) should behave

Verb	**Action**	**Idempotent?**	**Cacheable?**
`GET`	Retrieve data	**Yes**	**Yes**
`POST`	Create resource	No	No
`PUT`	Replace/Update	**Yes**	No
`PATCH`	Partial Update	No	No
`DELETE`	Remove resource	**Yes**	No

Status Codes as Metrics

- Are the equivalent of return values
- Allow to monitor when things go wrong

2xx Success → All good
4xx Client Error → Bad config, missing auth
5xx Server Error → App crashed, DB timed out
502/504 Upstream is down or the gateway timed out

The Reverse Proxy & Load Balancer

Entrypoint for distributed applications

Key Responsibilities

- **SSL Termination:** Offloading CPU-heavy crypto from app nodes
- **Path-Based Routing:** `/api` goes to Service A, `/static` to S3/CDN
- **Retries & Timeouts:** Handling "Partial Failures" in the distributed chain

L7 Health Checking

- When some services fail it is normally hard to know
- Monitoring (observability) is key in this environment

Layered health check layout

- **Basic:** L4 Check – Is the HTTP(S) port open?
- **Intermediate:** L7 Check – Does `GET /health` return `200 OK`?
- **Advanceck:** Deep Check – Does the app check its DB connection before responding?

Web Servers

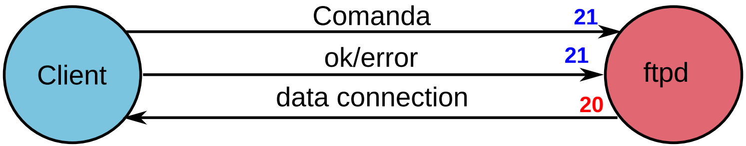
- Implement support for HTTP
- Most can act as proxies → but normally is better to separate the roles

Main features

- Unprivileged user execution
- Queries are served using memory separated processes/threads
  - Memory sharing configurable by the administrator
  - Maximum concurrent processes limit
- Configuration options in a per directory basis
- Virtual Host configuration
  - By IP address
  - By DNS name

File Transfer Protocol (FTP)

- Data transfer service
- Connection oriented
- Control connection
  - There is state between connections: `cwd`
- Data connection
  - **active**: does not support NAT
  - **passive**: NAT is supported
  - There is a new data connection per transfer

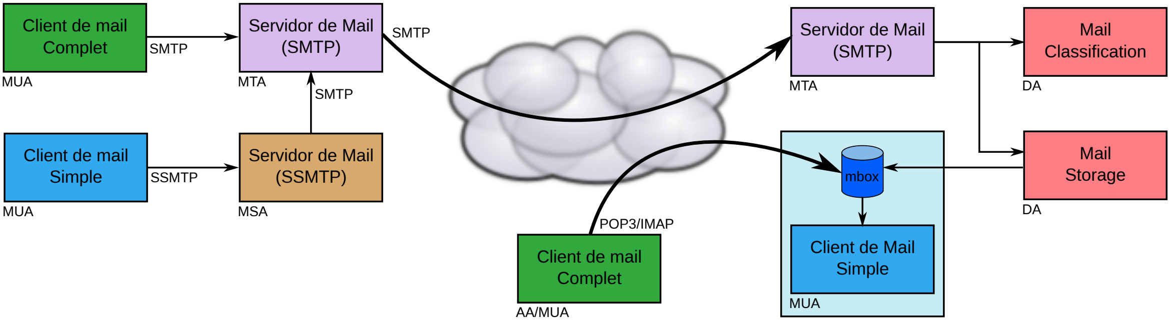


Simple Mail Transfer Protocol (SMTP)

Parts composing the mail system

- MUA - Mail User Agent
  - User application to read/write e-mails
- MSA - Mail Submission Agent
  - Application to transmit the mail from the client to the MTA
  - It make all previous error checking
- MTA - Mail Transport Agent
  - It sends the e-mail between servers
- Delivery Agent
  - Application to store mails into the user's mailbox
  - Sometimes the mails are stored into a database
- Access Agent
  - Application allowing the user to access its e-mail

Mail system components



Internals of an e-mail

- Envelope
  - Message destination
  - Source
  - Not received by the clients – only for servers
- Headers
  - Set of message properties
    - Sending date
    - Source and destination (shown by the e-mail clients)
    - List of servers the message has crossed
- Message body
  - Uses 7 bits ASCII
  - Attachments use Base-64

E-Mail server configuration

Mail sending – e.g. postfix

- Sending direct to the destination
  - Search for MX record in DNS – local destination
- Sending through a Relay
  - No direct access to the destination

Mail reception – e.g. dovecot

- Store the mails locally
  - POP3, DIMAP
- Store the mails in the remote server
  - IMAP

E-mail reception

Post Office Protocol (POP)

- It allows users to access their mailbox
- It downloads the messages to the local machine
- Authentication without encryption
  - `pop3s` secure alternative using SSL

Internet Message Access (IMAP)

- It allows users to manage their mailbox
- Management is performed remotely
- User authentication
  - Allowing encryption
- `imaps` even more secure alternative using SSL

Security considerations

User authentication

- By default the server does not ask for credentials
  - SASL can be used
- Envelope can be forged — SPAM . . .
- Trust mail relays
  - The server always tries to send the message
  - Even if the headers do not belong to the domain (Open Relays)

Security considerations

Mail privacy

- Mail is sent in plain text
  - Use of TLS (SSL) only between MUA and MTA
- PGP - Pretty Good Privacy
  - Message cyphering and signing
  - Based in public key cryptography
- S/MIME

Filter installation

- Anti-spam
  - `Spamassasin`, gray lists, black lists, ...
- Anti-virus
  - `Clam AV`, `Amavis`, `f-prot`,...

Security Considerations

To prevent spoofing, we use a layered approach:

- **SPF:** Who is allowed to send?
- **DKIM:** Is the email authentic and untampered?
- **DMARC:** What happens if the checks fail?

Sender Policy Framework (SPF)

SPF is a `DNS TXT` record

Example SPF Record

```
v=spf1 ip4:192.168.1.1 include:_spf.google.com ~all
```

- **ip4/include:** Authorized sources
- **~all:** "Soft fail" (mark as suspicious)
- **-all:** "Hard fail" (reject outright)

DKIM: DomainKeys Identified Mail

DomainKeys Identified Mail (DKIM) provides cryptographic proof that the mail was authorized by the domain owner

- **How it works:** The sender signs the email with a **private key**. The receiver verifies it using a **public key** found in DNS
- **Integrity:** It hashes the body and specific headers. If a single character is changed in transit, the signature breaks

The DKIM-Signature Header (Simplified)

```
v=1; a=rsa-sha256; d=example.com; s=jan2026;
bh=Zkw9...  (Body Hash);
b=mS92...  (Digital Signature)
```

- **Selector (s=):** Allows a domain to have multiple keys (e.g., one for Marketing, one for Corporate)

Domain-based Message Authentication, Reporting & Conformance (DMARC)

DMARC ensures alignment between the "Header From" address and SPF/DKIM checks.

Policy	Action
p=none	Monitor only; no impact on delivery.
p=quarantine	Send failed mail to Spam.
p=reject	Bounce the email entirely.

Exercise – In group

We just set up a filter to control spam

- Which action would you take as a server when you detect a spam message?
- And if the filter is an anti-virus?

Secure Shell

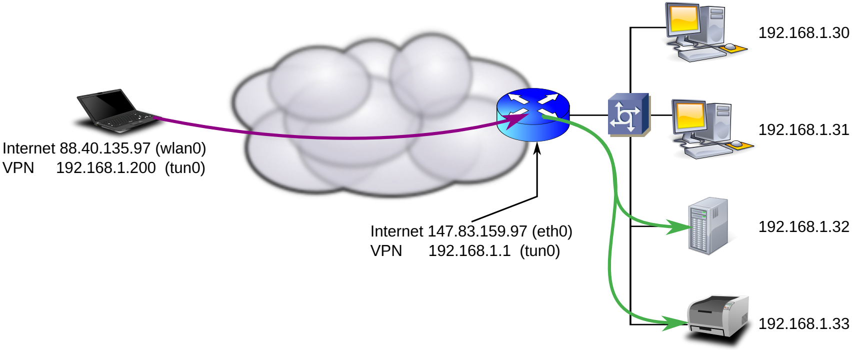
- It substitutes rsh/rlogin and telnet
- Adding security
  - It performs authentication based on RSA, DSA, ECDSA
    - Session key is signed by the client's private key
    - The server uses the public key as stored in (`.ssh/authorized_keys`) to check if the signature is correct
    - password based authentication is also supported
  - Connection is fully encrypted
    - Confidentiality: `3DES`, `Blowfish`, `...`
    - Integrity: `hmac-md5`, `...`
- The server runs the specified command or offer a shell
- Transparent session
  - Whenever a pseudo-terminal is not requested
  - It can be used to transfer binary files

Lightweight Directory Access Protocol (LDAP)

- It provides access to users database
  - Directory format (X.500)
- It offers user authentication methods
  - `/etc/passwd`, `/etc/shadow`, `/etc/group`, . . .
  - . . . they can be dumped to the LDAP database
- Besides regular files, login can also be controlled through the database
- It is used extensively on Windows Server Active Directory

Virtual Private Networks (VPN)

- Server and client negotiate a secure connection
- An internal IP is offered through a secure tunnel
  - It grants access to all the internal services

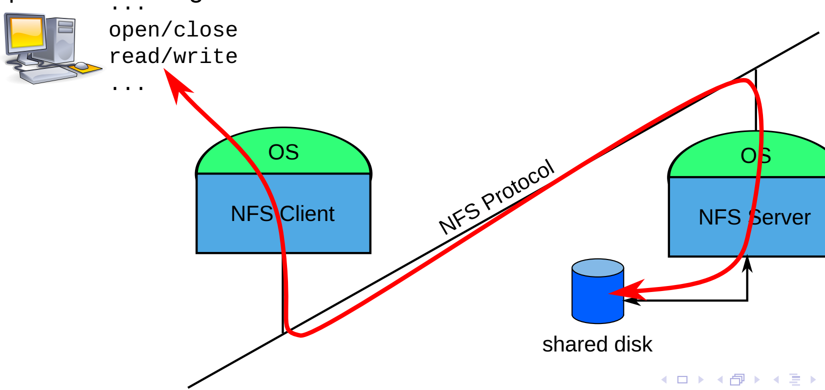


Outline

1. Introduction
2. Servers
3. Service Brokers
4. Pure services
5. Network file Sharing
   - Network File System (NFS)
   - Samba (SMB)
6. Monitoring

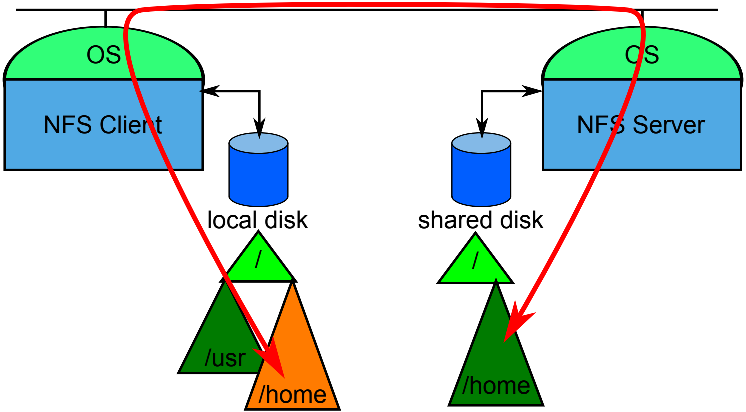
Network File System (NFS)

- File access in a remote server
  - Keeping the semantics (privilege wise) of the local filesystem
- It is transparent to the user
  - Implemented using RPC's



Remote mounting for NFS

The mounted directory is presented as local



Access privileges

- UIDs in the remote machines must be the same as used in local
  - Filesystems store UID rather than usernames
  - This can be adapted by using `idmapd`
- UID automatic translation (`idmapd`)
  - `root`, `nobody`
- Options
  - `no_root_squash`, root can su to any user!
  - `all_squash`, all users become `nobody`
  - We can decide who nobody is

```
anonuid=UID,anongid=GID
```

NFS Configuration

- Determine which resources to export
- Hosts to export to
- Configuration flags

/etc/exports

```
/              master(rw) trusty(rw,no_root_squash)
/projects      proj*.local.domain(rw)
/usr           *.local.domain(ro) @trustedgroup(rw)
/home/joe      pc001(rw,all_squash,anonuid=150,anongid=100)
/pub           (ro,insecure,all_squash)
```

SMB — Samba

- It allows sharing files and printers
- User level access control
  - Authentication using login and password
    - Based on username not UID
    - Encripted and plaintext password transmission
  - Machine based access restriction
    - It does not allow to change permissions depending on the source
    - One must use different share names

```
[global]
        workgroup = ASOWG
        server string = %h (Samba, Debian)
        server min protocol = SMB2_10
        interfaces = 127.0.0.0/8 virbr0
        max log size = 1000
        server role = standalone server
        obey pam restrictions = yes
        unix password sync = yes
        passwd program = /usr/bin/passwd %u

[homes]
        comment = Home Directories
        browseable = yes
        read only = no
        create mask = 0700
```

```
[printers]
        comment = All Printers
        browseable = no
        printable = yes
        guest ok = no
        read only = yes
        create mask = 0700
[print$]
        comment = Printer Drivers
        path = /var/lib/samba/printers
        browseable = yes
        read only = yes
        guest ok = no
```

Outline

Packet Sniffing — tcpdump

```
aso@localhost:~$ sudo tcpdump -s 1024 -X -c 10 -ni any icmp
tcpdump: WARNING: any: That device doesn't support promiscuous mode
(Promiscuous mode not supported on the "any" device)
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 1024 bytes
23:00:11.427594 wlan0 Out IP 192.168.1.172 > 147.83.2.135: ICMP echo request, id 3789, seq 1, length 64
0x0000:  4500 0054 9383 4000 4001 b4f6 c0a8 9bac  E..T..@.@.......
0x0010:  9353 0287 0800 44db 0ecd 0001 db93 e669  .S....D........i
0x0020:  0000 0000 1d86 0600 0000 0000 1011 1213  ................
0x0030:  1415 1617 1819 1a1b 1c1d 1e1f 2021 2223  .............!"#
0x0040:  2425 2627 2829 2a2b 2c2d 2e2f 3031 3233  $%&'()*+,-./0123
0x0050:  3435 3637                                4567
23:00:11.431676 wlan0 In  IP 147.83.2.135 > 192.168.1.172: ICMP echo reply, id 3789, seq 1, length 64
0x0000:  4500 0054 f245 4000 f301 a333 9353 0287  E..T.E@....3.S..
0x0010:  c0a8 9bac 0000 4cdb 0ecd 0001 db93 e669  ......L........i
0x0020:  0000 0000 1d86 0600 0000 0000 1011 1213  ................
0x0030:  1415 1617 1819 1a1b 1c1d 1e1f 2021 2223  .............!"#
0x0040:  2425 2627 2829 2a2b 2c2d 2e2f 3031 3233  $%&'()*+,-./0123
0x0050:  3435 3637                                4567
23:00:12.429103 wlan0 Out IP 192.168.1.172 > 147.83.2.135: ICMP echo request, id 3789, seq 2, length 64
0x0000:  4500 0054 969a 4000 4001 b1df c0a8 9bac  E..T..@.@.......
0x0010:  9353 0287 0800 84d4 0ecd 0002 dc93 e669  .S.............i
0x0020:  0000 0000 dc8b 0600 0000 0000 1011 1213  ................
0x0030:  1415 1617 1819 1a1b 1c1d 1e1f 2021 2223  .............!"#
0x0040:  2425 2627 2829 2a2b 2c2d 2e2f 3031 3233  $%&'()*+,-./0123
0x0050:  3435 3637                                4567
```

Service Detection—ss

Syntax

- `ss [options]`
- `-a` – Display both listening and non-listening (for TCP this means established connections) sockets.

```
aso@localhost:~$ ss -a
Netid State      Recv-Q  Send-Q               Local Address:Port      Peer Address:Port   Process
u_str ESTAB      0       0      /run/systemd/journal/stdout 40159                  * 38282
tcp   LISTEN     0       50                 0.0.0.0:bacula-fd      0.0.0.0:*
```

Service Detection—nmap

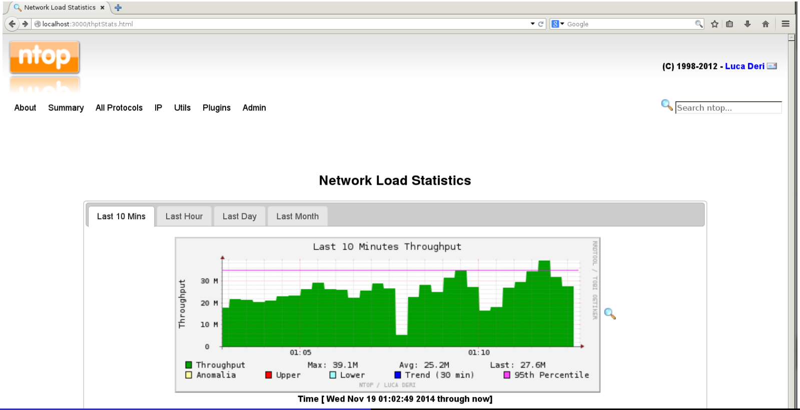
Syntax

- `nmap [options] IP_list`

```
aso@localhost:~$ nmap 192.168.1.2
Starting Nmap 6.47 ( http://nmap.org ) at 2014-11-19 00:18 CET
Nmap scan report for 192.168.1.2
Host is up (0.057s latency).
Not shown: 988 closed ports
PORT      STATE SERVICE
22/tcp    open  ssh
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
143/tcp   open  imap
443/tcp   open  https
514/tcp   open  shell
993/tcp   open  imaps
2049/tcp open  nfs
6566/tcp open  sane-port
9101/tcp open  jetdirect
9103/tcp open  jetdirect

Nmap done: 1 IP address (1 host up) scanned in 3.36 seconds
```

Other Applications

- suricata - Intrusion detection system
- Grafana Locki - Log Watcher
- ntop NG - Network Top



Outline

Task

A company has the following characteristics

- Company Executive Management has 5 PC.
- Administration department has 10 PC.
- Available IP addresses: 180.45.23.0/28
- The company needs the following services:
    - Web – General to the whole company
    - E-Mail – General to the whole company
    - File Sharing using NFS – Per department
    - VPN - General to the whole company
    - SSH - Present in all servers
    - DHCP
    - DNS - Server for the employees
    - Printing Service
    - HTTPS Intranet

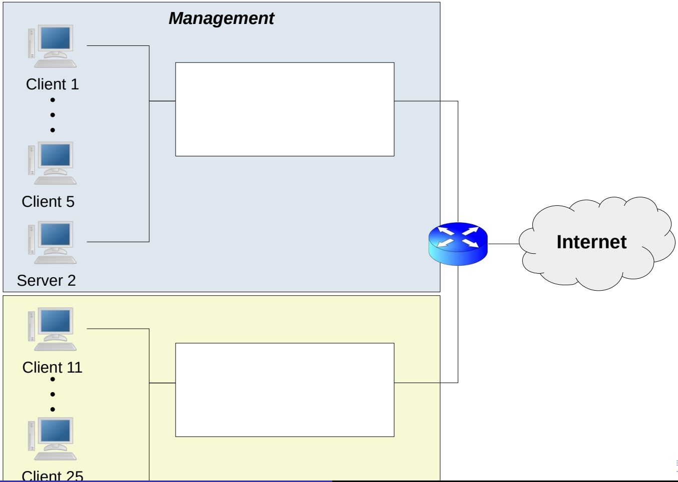
Task

Service Load

- Web Very High
- E-Mail High
- File Sharing using NFS Very High
- VPN Very Low
- SSH Very Low
- DHCP Low
- DNS Normal
- Printing Service Very Low
- HTTPS Intranet Normal

Task

Add all the necessary servers and network equipment



Task

Questions

- Would you buy more hardware
- Distribute all the services among the different servers
- Specify where would you install the firewall and its basic configuration (This will be done in lesson 9)